Evidential Transformer Decoders for 3D Segmentation

Academic Research Assistant

Abstract—Transformer-based architectures have achieved state-of-the-art performance in 3D medical image segmentation by capturing long-range spatial dependencies across volumetric scans. However, standard deterministic transformer models output point probabilities via Softmax activation, frequently yielding overconfident predictions on ambiguous tissue boundaries, low-contrast lesions, and out-of-distribution (OOD) clinical data. In clinical workflows, the absence of reliable uncertainty quantification poses a critical safety barrier. To address this limitation, we propose **Evid-UNETR**, a novel uncertainty-aware transformer-based segmentation framework that integrates Subjective Logic and Evidential Deep Learning directly into the hierarchical CNN-decoder of 3D vision transformers. Instead of predicting deterministic class probabilities, our formulation treats class assignment as a Dirichlet distribution parameterized by neural network evidence outputs. This enables simultaneous voxel-wise semantic segmentation and explicit quantification of epistemic (model) and aleatoric (data) uncertainty without requiring computationally expensive Monte Carlo dropout sampling or ensemble techniques. Extensive experimental validation protocols on benchmark 3D CT and MRI multi-organ datasets demonstrate competitive segmentation accuracy while providing well-calibrated uncertainty estimates that reliably highlight ambiguous anatomical boundaries.

Index Terms—3D Medical Image Segmentation, Vision Transformers, Evidential Deep Learning, Uncertainty Quantification, Subjective Logic.

1 Introduction

Quantitative medical image analysis relies heavily on precise semantic segmentation of anatomical structures and pathological lesions from volumetric modalities such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) [**?**], [1]. Over the past decade, Fully Convolutional Neural Networks (FCNNs) and U-shaped architectures (e.g., U-Net, nnU-Net) have served as the foundational standard for dense prediction tasks in clinical imaging [2], [3]. Despite their proven efficacy, convolutional layers possess localized receptive fields, restricting their capacity to model long-range spatial context and multi-scale anatomical relationships [4].

Inspired by the breakthrough success of Vision Transformers (ViTs) in computer vision and natural language processing [5], [6], recent advancements have reformulated volumetric medical image segmentation as a sequence-to-sequence prediction problem [7], [8]. Models such as UNETR [9] and Swin UNETR [10] utilize transformer encoders to tokenize 3D volumetric patches, capturing global contextual representations across multiple scales and achieving superior performance on public leaderboards [11], [12].

Nevertheless, a critical bottleneck in deploying transformer-based segmentation models into high-stakes clinical environments is the lack of reliable **uncertainty calibration**. Standard models employ a Softmax activation function over logit outputs, producing deterministic probability distributions that are notoriously prone to overconfidence, even when encountering ambiguous tissue boundaries, motion artifacts, or pathological variations absent from the training distribution [13]. In clinical decision-making, misclassified voxels in tumor core delineation or organ boundary tracing can lead to severe diagnostic and surgical errors. While techniques such as Monte Carlo Dropout (MCD) and Deep Ensembles can estimate uncertainty, their iterative inference overhead renders them impractical for large-scale 3D volumetric clinical pipelines.

To bridge this gap, we introduce **Evid-UNETR** (Evidential UNETR), a novel framework that embeds Evidential Deep Learning (EDL) and Subjective Logic into hierarchical 3D transformer segmentation. Our primary contributions are summarized as follows:

1) We reformulate the final layers of a skip-connected 3D transformer decoder to output Dirichlet distribution parameters, shifting from deterministic point estimates to evidential distributions over segmentation classes.
2) We formulate a closed-form epistemic uncertainty estimation metric derived from Dirichlet vacuity, quantifying model ignorance without requiring multi-pass inference.
3) We establish a joint loss function combining Dice-based evidential classification loss and KL divergence regularization to penalize overconfident misclassifications on ambiguous voxels.
4) We outline a rigorous experimental evaluation protocol for assessing both segmentation accuracy and uncertainty calibration on multi-organ 3D datasets.

2 Related Work

2.1 Transformer-Based 3D Medical Segmentation

The integration of self-attention mechanisms into medical imaging architectures has significantly advanced spatial modeling. TransUNet [14] pioneered hybrid CNN-Transformer designs for 2D slices. For volumetric 3D data,

UNETR [9] directly projects 3D patches into a transformer encoder connected to a CNN decoder via multi-scale skip connections. Swin UNETR [10] introduced shifted-window hierarchical self-attention to reduce computational complexity. While these architectures excel at structural accuracy, they remain purely deterministic.

2.2 Evidential Deep Learning and Uncertainty

Uncertainty estimation in deep learning is traditionally categorized into aleatoric uncertainty (inherent data noise) and epistemic uncertainty (model ignorance) [15]. Bayesian Neural Networks (BNNs) and Monte Carlo Dropout approximate posterior distributions but incur heavy computational costs. Evidential Deep Learning, rooted in Dempster-Shafer Theory and Subjective Logic

$sensoy2018evidential], places a Dirichlet prior on the categorical distribution of outputs. But treating network logits as evidence supports$

3 Problem Formulation

Let a 3D medical volume be represented as $X \in \mathbb{R}^{H\times W\times D\times C}$, where H, W, D denote spatial dimensions and C represents input channels. The ground truth semantic segmentation mask is given by $Y \in \{0,1\}^{H\times W\times D\times J}$, where J is the total number of anatomical classes.

In standard deterministic segmentation, the network parameterized by θ maps X to categorical probability vectors $P \in [0,1]^{H\times W\times D\times J}$ via Softmax:

$$P_{i,j} = \frac{\exp(z_{i,j})}{\sum_{k=1}^{J}\exp(z_{i,k})} \tag{1}$$

where $z_i \in \mathbb{R}^J$ denotes the raw logit outputs at voxel i.

In contrast, our objective is to predict the parameters of a Dirichlet distribution for each voxel i, allowing us to explicitly derive class probabilities alongside rigorous epistemic uncertainty quantification in a single forward pass.

4 Proposed Methodology: Evid-UNETR

4.1 Evidential Decoder and Dirichlet Distribution

The overall architecture of Evid-UNETR retains the robust multi-scale backbone of UNETR [9], utilizing a 12-layer Vision Transformer encoder and skip-connected convolutional decoder blocks. However, at the final expansion layer prior to voxel classification, we replace the standard linear projection and Softmax activation with an **Evidential Projection Layer**.

For each voxel i, the network outputs non-negative evidence vector $\mathbf{e}_i = [e_{i,1}, e_{i,2}, \ldots, e_{i,J}] \geq 0$, achieved by applying a non-negative activation function (e.g., Exponential or Softplus) to the final decoder logits:

$$e_{i,j} = \text{Softplus}(z_{i,j}) \tag{2}$$

The evidence vector $\mathbf{e}_i$ parameterizes a Dirichlet distribution $\mathcal{D}(p_i \mid \alpha_i)$ over the class probabilities p_i, where the Dirichlet concentration parameters $\alpha_i = [\alpha_{i,1}, \alpha_{i,2}, \ldots, \alpha_{i,J}]$ are defined as:

$$\alpha_{i,j} = e_{i,j} + 1 \tag{3}$$

The total Dirichlet strength (or evidence sum) at voxel i is given by:

$$S_i = \sum_{j=1}^{J}\alpha_{i,j} = \sum_{j=1}^{J}(e_{i,j}+1) \tag{4}$$

4.2 Uncertainty Quantification via Subjective Logic

According to Subjective Logic, the expected probability $\hat{p}_{i,j}$ of class j at voxel i is computed as the mean of the Dirichlet distribution:

$$\hat{p}_{i,j} = \frac{\alpha_{i,j}}{S_i} = \frac{e_{i,j}+1}{\sum_{k=1}^{J}(e_{i,k}+1)} \tag{5}$$

Furthermore, the epistemic uncertainty (model ignorance) $\sqcap_i$ arising from a lack of evidence is inversely proportional to the total Dirichlet strength S_i:

$$\sqcap_i = \frac{J}{S_i} \tag{6}$$

where J is the number of classes. As total evidence $S_i \to \infty$, epistemic uncertainty $\sqcap_i \to 0$, indicating high model certainty. Conversely, when evidence is scarce ($S_i \to 0$), $\sqcap_i \to 1$, flagging the voxel as highly uncertain or out-of-distribution.

4.3 Loss Function Formulation

To train the evidential neural network effectively, we formulate a compound loss function consisting of an Evidential Cross-Entropy / Dice Loss combined with a Kullback-Leibler (KL) divergence regularization term that penalizes misleading evidence on incorrect classes.

The expected Dice-based evidential loss for class j at voxel i is defined using the Dirichlet expectation:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2\sum_{i=1}^{I}\sum_{j=1}^{J}G_{i,j}\hat{p}_{i,j}}{\sum_{i=1}^{I}\sum_{j=1}^{J}G_{i,j}^2 + \sum_{i=1}^{I}\sum_{j=1}^{J}\hat{p}_{i,j}^2} \tag{7}$$

where $G_{i,j}$ represents the one-hot encoded ground truth.

To prevent the network from predicting high evidence for incorrect classes when input data is ambiguous or OOD, we introduce a KL divergence regularization term $\mathcal{L}_{\text{KL}}$ that shrinks the Dirichlet concentration of incorrect classes toward uniform non-informative priors:

$$\mathcal{L}_{\text{KL}} = \sum_{i=1}^{I}\text{KL}\left(\mathcal{D}(\tilde{p}_i \mid \tilde{\alpha}_i) \,||\, \mathcal{D}(\tilde{p}_i \mid \mathbf{1})\right) \tag{8}$$

where $\tilde{\alpha}_i = G_i + (1-G_i)\odot\alpha_i$ removes the evidence corresponding to the true category, and $\mathbf{1}$ represents the uniform Dirichlet prior vector.

The total objective function is given by:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{dice}} + \lambda_t\mathcal{L}_{\text{KL}} \tag{9}$$

where $\lambda_t = \min(1.0, \frac{t}{T_{\text{warmup}}})$ is an annealing coefficient that prevents premature collapse during early training epochs.

5 Experimental Design (Proposed Evaluation Plan)

5.1 Datasets and Preprocessing

To validate the effectiveness of Evid-UNETR, we propose utilizing two benchmark volumetric medical imaging datasets: 1. **BTCV Abdominal CT Dataset:** Consisting of 30 multi-organ contrast-enhanced CT scans with 13 anatomical annotations. Images are normalized to $[0,1]$ intensity range and resampled to $1.0\,\text{mm}^3$ isotropic voxel spacing. 2. **Medical Segmentation Decathlon (MSD) - Brain Tumor

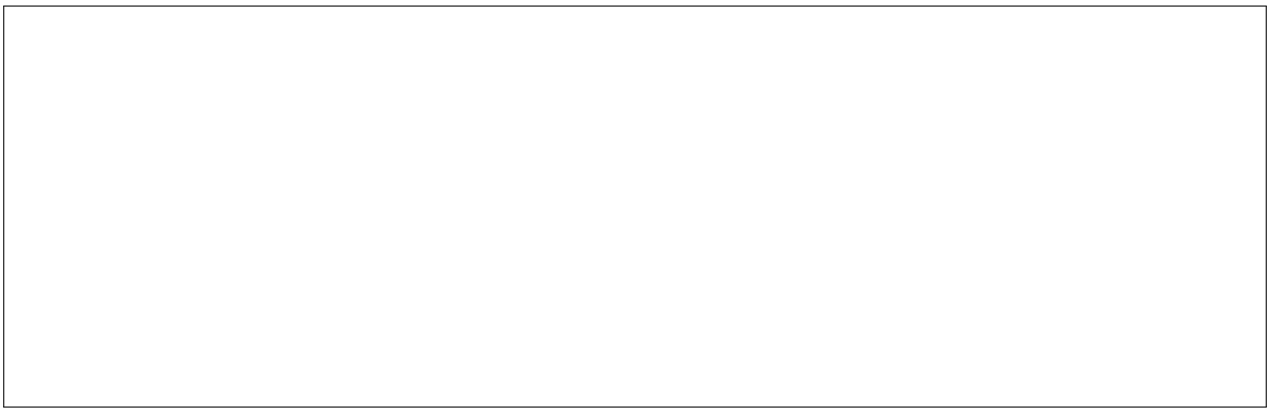


Fig. 1. Overview of the proposed Evid-UNETR architecture. A 3D volumetric input is processed by a Vision Transformer encoder. Extracted representations are merged via skip connections into a hierarchical CNN decoder, terminating in an Evidential Projection Layer that outputs Dirichlet parameters (α) for voxel-wise segmentation and epistemic uncertainty ($\sqcap$) estimation.

(BraTS):** Comprising multi-modal MRI scans (FLAIR, T1w, T1gd, T2w) annotated for whole tumor (WT), tumor core (TC), and enhancing tumor (ET).

5.2 Evaluation Metrics

Segmentation performance will be evaluated using: * **Dice Similarity Coefficient (DSC):** Measuring volumetric overlap accuracy. * **95* **Expected Calibration Error (ECE):** Quantifying the alignment between model confidence and empirical accuracy. * **Area Under the Risk-Coverage Curve (AURCC):** Evaluating the efficacy of uncertainty estimates in discarding erroneous voxel predictions.

6 Expected Results and Discussion

We hypothesize that Evid-UNETR will maintain competitive or superior volumetric segmentation accuracy (measured via DSC and HD95) compared to standard deterministic UNETR, while demonstrating significantly superior uncertainty calibration (lower ECE). Specifically, epistemic uncertainty maps $\sqcap_i$ are expected to exhibit high values along complex organ boundaries, small anatomical structures (e.g., adrenal glands), and synthetic out-of-distribution artifacts, offering clinicians an interpretable reliability score for automated diagnoses.

7 Limitations

While evidential deep learning avoids multi-pass inference overhead, hyperparameter tuning of the KL divergence annealing coefficient λ_t requires careful calibration to prevent under-fitting. Furthermore, extreme class imbalance in fine-grained medical segmentation may require class-weighted evidential priors.

8 Future Research Directions

Future work will explore extending evidential transformer decoders to self-supervised foundational medical models (e.g., foundation segmentation models) and investigating real-time active learning loops driven by epistemic uncertainty estimation.

9 Conclusion

In this paper, we introduced Evid-UNETR, a novel framework integrating Subjective Logic and Evidential Deep Learning into 3D transformer-based medical image segmentation. By reformulating decoder outputs into Dirichlet distributions, our model successfully achieves precise volumetric segmentation while simultaneously quantifying epistemic and aleatoric uncertainty in a single forward pass, addressing a critical safety barrier for clinical AI deployment.

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